

MATH I - Week 1

Fall 2026

Practical sessions material

Structure of the Practical sessions:

- 1. Practical sessions can start with questions of students.
 - 2. Then, exercises will be presented for students to solve (students can try, for a few minutes to solve them on their notebook/paper or speak answers)
 - 3. Solutions will be presented (after each exercise)
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Exercise 1: Decimal representations

Classify each decimal as terminating, repeating or nonrepeating. State whether the corresponding number is rational or irrational.

1. 0.375
 2. 0.272727 ...
 3. 1.101001000100001 ...
 4. 3.14159265 ...
 5. 0.123123123 ...
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Exercise 2: Intervals

Write each set using interval notation.

1. $x \in \mathbb{R} : -3 \leq x < 4$
2. $x \in \mathbb{R} : x \neq 3$

Then represent each set on the real number line.

Exercise 3: From intervals to inequalities

Write each interval using inequalities.

- a. $[-2, 5)$, b. $(-\infty, 4]$, c. $(1, \infty)$,
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...About Irrational Numbers (to remind us)...

The set of irrational numbers is

$$\mathbb{R} \setminus \mathbb{Q}.$$

An irrational number cannot be expressed as

$$\frac{p}{q}, \quad p, q \in \mathbb{Z}, \quad q \neq 0.$$

Exercise 4: Rational or irrational?

Classify each number as rational or irrational.

a. $\sqrt{3}$, b. $\sqrt{25}$, c. $\sqrt{50}$,

i. $(1 + \sqrt{5})(1 - \sqrt{5})$, j. $\frac{1}{\sqrt{3}}$.

...Complex numbers...

Exercise 5: Basic complex arithmetic

Given

$$z_1 = 3 + 2i, \quad z_2 = 1 - 4i,$$

calculate:

$$z_1 + z_2, \quad z_1 - z_2, \quad z_1 z_2, \quad \frac{z_1}{z_2}.$$

Write every answer in the form $a + bi$.

Exercise 6: Complex conjugate and modulus

For each number, find its conjugate and modulus.

$$3 + 4i, \quad -2 + 5i, \quad 6i, \quad -7.$$

[Optional] Verify for each example that $z\bar{z} = |z|^2$.

Exercise 7: Complex division

Express the result in algebraic form $a + bi$.

$$\frac{1}{1+i}$$

Exercise 8: Quadratic equations

Solve over $\mathbb{C}$:

$$x^2 + 9 = 0$$

Exercise 9: Complex plane

Represent the following number in the complex plane:

$$2 + 3i$$

Determine:

1. its real part;
2. its imaginary part;
3. its modulus;
4. the quadrant in which it lies.

What will change if the number is

$$2 - 3i$$

Exercise 10: Polar form

Write the following complex number in trigonometric form $z = r(\cos \theta + i \sin \theta)$ and exponential form $z = re^{i\theta}$.

$$1 + i$$

Give θ in radians.

Exercise 11: de Moivre's formula

Use

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos(n\theta) + i \sin(n\theta))$$

to calculate:

$$(\sqrt{3} + i)^4$$

Check the result by algebraic multiplication.

Exercise 12: Roots of a complex number

Find the solution and represent it in the complex plane.

$$z^3 = -8.$$
